

Computational Statistics - STAT 575 - HW #1

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Problem 1

Write your own code and find solution to the equation $x^3 + x - 4 = 0$ using Newton's method and the secant method. Compare the number of iterations needed for different starting values for the two methods.

Solution: Newton's method

If we have some function $f : \mathbb{R} \mapsto \mathbb{R}$ and we wish to find a root of f , i.e., an x such that $f(x) = 0$, we may use Newton's method.

We take an initial guess of the root as x_0 and try to refine it with a linear approximation of f given by

$$L(f|x_0) := \lambda x. f(x_0) + f'(x_0)(x - x_0).$$

Now, we may approximate a root of f with a root of $L(f|x_0)$,

$$L(f|x_0)(x) = 0,$$

which may be rewritten as

$$f(x_0) + f'(x_0)(x - x_0) = 0.$$

Solving for a root x of $L(f|x_0)$ we get the result

$$x = x_0 - \frac{f(x_0)}{f'(x_0)}.$$

Hoping that x results in a better approximation of the root of f than x_0 , we approximate f with $L(f|x)$ and repeat the process.

Generalizing this result, we obtain the iterative procedure

$$x_{i+1} = x_i - \frac{f(x_i)}{f'(x_i)}.$$

We continue this process until we obtain some stopping condition, e.g., $|x_{i+1} - x_i| < \epsilon$.

Letting $f(x) := x^3 + x - 4$ and $f'(x) = 3x^2 + 1$ and substituting into the above result, we get the result

$$x_{i+1} = x_i - \frac{x_i^3 + x_i - 4}{3x_i^2 + 1}.$$

We implement a general procedure for Newton's method:

```
newton_method <- function(f,dfdx,x0,eps,debug=T)
{
  n <- 0
  repeat
```

```

{
  x1 <- x0 - f(x0) / dfdx(x0)
  n <- n + 1

  if (debug==T) { cat("iteration=",n," x=",x1,"\n") }
  if(abs(x1 - x0) < eps)
  {
    break
  }
  x0 <- x1
}

list(root=x0,iter=n)
}

```

We take an initial guess of $x_0 = 1$ and $\epsilon = 1 \times 10^{-6}$ and run the following R code to solve for a root of f using Newton's method:

```

f <- function(x) { x^3 + x - 4 }
dfdx <- function(x) { 3*x^2 + 1 }
eps <- 1e-6
x0 <- 1
result <- newton_method(f,dfdx,x0,eps)

```

```

## iteration= 1  x= 1.5
## iteration= 2  x= 1.387097
## iteration= 3  x= 1.378839
## iteration= 4  x= 1.378797
## iteration= 5  x= 1.378797

```

We obtain $x \approx 1.3787967$ after 5 iterations. When we plug that approximate root into f we obtain the result $f(1.3787967) = 7.3825959 \times 10^{-9}$, which is approximately zero.

Solution: Secant method

In Newton's method, we linearize f using the derivative of f . If, instead, we use the secant of f with respect to two inputs x_i and x_{i+1} , as given by

$$\frac{f(x_{i+1}) - f(x_i)}{x_{i+1} - x_i},$$

we get the iterative procedure

$$x_{i+2} = x_{i+1} - f(x_{i+1}) \frac{x_{i+1} - x_i}{f(x_{i+1}) - f(x_i)},$$

which requires two initial values x_0 and x_1 .

We define the secant method as a function given by:

```

secant_method <- function(f,x0,x1,eps,debug=T)
{
  n <- 0
  repeat
  {
    x2 <- x1 - f(x1) * (x1 - x0) / (f(x1) - f(x0))
    n <- n + 1

    if (debug==T) { cat("iteration=",n," x=",x2,"\n") }
  }
}

```

```

    if(abs(x2-x1) < eps)
    {
        break
    }
    x0 <- x1
    x1 <- x2
}

list(root=x1,iter=n)
}

```

We let $x_0 = 0$, $x_1 = 1$, and keep everything else the same and run the secant method with the following R code:

```

x0 <- 0
x1 <- 1
result <- secant_method(f,x0,x1,eps,F)

```

We obtain $x \approx 1.3787965$ after 7 iterations. When we plug that approximate root into f we obtain the result $f(1.3787965) = -1.268648 \times 10^{-6}$, which is approximately zero.

Note that this is 2 more iterations than Newton's method.

Comparison of Newton's method versus secant method

We perform 10000 trials to get a better view of how the two methods, Newton and secant, compare over many different initial guesses.

We generate the data with:

```

n <- 10000
from <- 0
to <- 4
by <- (to-from)/n
newt_sols <- vector(length=n)
sec_sols <- vector(length=n)
i <- 1
for (x0 in seq(from=from, to=to, by=by))
{
    newt_sols[i] <- newton_method(f,dfdx,x0,eps,F)$iter
    sec_sols[i] <- secant_method(f,x0,x0+1,eps,F)$iter
    i <- i + 1
}

```

We summarize the results and report them with:

```

cat("mean iterations\n",
    "newton => ", mean(newt_sols), "\n",
    "secant => ", mean(sec_sols), "\n")

```

```

## mean iterations
## newton => 5.819618
## secant => 7.231077

```

We see that Newton's method, on average, requires 1.4114589 fewer iterations before the stopping condition is satisfied.

Problem 2

Poisson regression. The Ache hunting data set has $n = 47$ observations recording is the number of monkeys killed over a period of days with each hunter along with hunter's age. It is of interest to estimate and quantify the monkey kill rate as a function of hunter's age. Hunting prowess confers elevated status among the group, so a natural question is whether hunting ability improves with age, and at which age hunting ability is best.

Hand-code Newton-Raphson in R to fit the Poisson regression model

$$\text{monkeys}_i \sim \text{Pois}(\exp(\log \text{days}_i + \theta_1 + \theta_2 \text{age}_i + \theta_3 \text{age}_i^2)).$$

Feel free to use jacobian and hessian in the numDeriv R package. You may need a sets of crude starting values. I run a linear regression for the “empirical log-rates” and get starting values (5.99, 0.167, 0.001). Feel free to use those. Compare your result with glm() function in R using

```
glm(monkeys~age+I(age^2), family="poisson", offset=log(days), data=d)
```

Solution

We are given the following data:

```
d <- read.table("ache.txt", header=T)
n <- length(d$age)
X <- cbind(rep(1,n), d$age, d$age^2)

loglike <- function(theta)
{
  sum(dpois(d$monkeys,exp(log(d$days)+X%*%theta),log=T))
}
```

We generalize the univariate Newton's method in Problem 1 to the multivariate case. We implement the multivariate Newton-Raphson method with numerical hessian and jacobian with the following R code:

```
library(numDeriv)
newton_raphson_method <- function(x0,f,eps)
{
  n <- 0
  x1 <- x0
  repeat
  {
    x1 <- x0 - solve(hessian(f,x0))%*%t(jacobian(f,x0))
    n <- n + 1
    if (n %% 7 == 0) { cat("iteration=",n," theta=",x1,"\n") }
    if (max(abs(x1 - x0)) < eps)
    {
      break
    }
    x0 <- x1
  }
  list(root=x1,iter=n)
}
```

We use the multivariate Newton-Raphson method to find the MLE of θ in the poisson regression model:

```
eps <- 1e-6
theta0 <- c(5.99, 0.167, 0.001) # starting values
theta_mle <- newton_raphson_method(theta0,loglike,eps)$root
```

```
## iteration= 7  theta= -1.003658 0.1667888 0.001001747
## iteration= 14  theta= -7.58365 0.1530251 0.00111402
## iteration= 21  theta= 1.438523 -0.2696733 0.003827107
## iteration= 28  theta= -5.484246 0.1246477 -0.001203418
```

The MLE of θ is given by:

```
theta_mle
```

```
##           [,1]
## [1,] -5.484245904
## [2,]  0.124647667
## [3,] -0.001203418
```

We compare the results with the builtin method:

```
glm(monkeys~age+I(age^2),family="poisson", offset=log(days),data=d)$coefficients
```

```
## (Intercept)      age      I(age^2)
## -5.484245904  0.124647667 -0.001203418
```

The hand-coded approach and the builtin approach obtain the same point estimate $\hat{\theta} = (-5.4842, 0.1246, -0.0012)'$.

Problem 3

Logistic and Cauchy distributions are well-suited to the inverse transform method. For each of the following, generate 10,000 random variables using the inverse transform. Compare your program with the built-in R functions `rlogis()` and `rcauchy()`, respectively:

Solution: part (a)

Standard logistic distribution

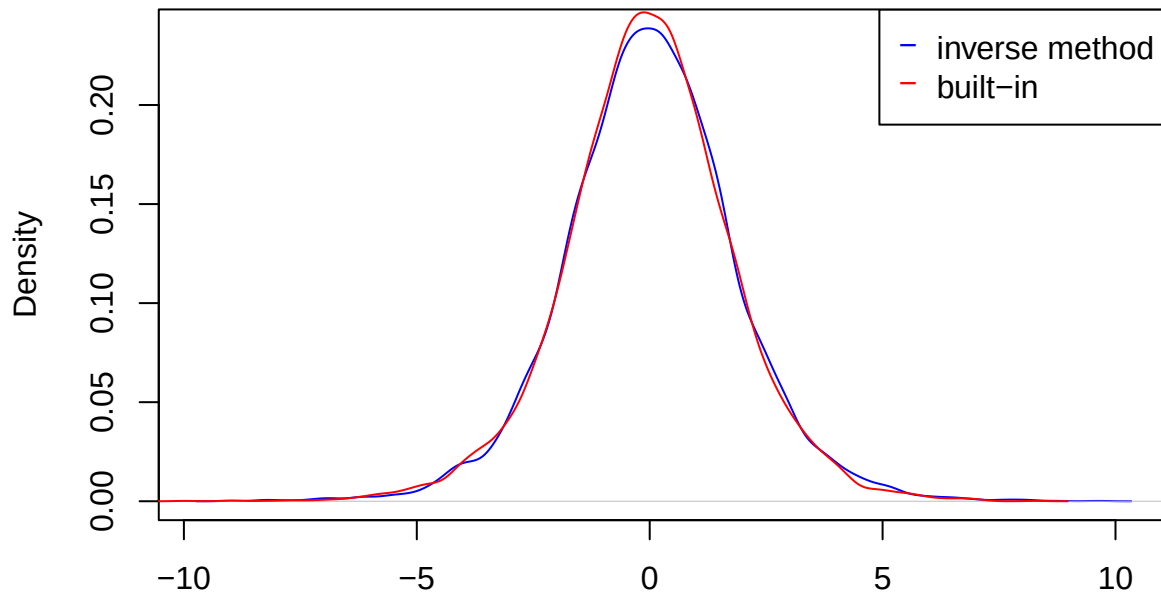
$$F(x) = \frac{1}{1 + e^{-x}}$$

Solve for x in

$$\begin{aligned} u &= F(x) \\ u &= \frac{1}{1 + e^{-x}} \\ x &= \log(u/(1 - u)). \end{aligned}$$

```
n <- 10000
us <- runif(n)
d1 <- density(log(us/(1-us)))
d2 <- density(rlogis(n))
plot(d1,col="blue",main="comparison of density plots")
lines(d2,col="red")
legend(x="topright",legend=c("inverse method","built-in"),col=c("blue","red"),
      pch=c("-", "-"))
```

comparison of density plots



N = 10000 Bandwidth = 0.239

Solution: part (b)

Standard Cauchy distribution

$$F(x) = \frac{1}{2} + \frac{1}{\pi} \arctan(x)$$

Solve for x in

$$\begin{aligned} u &= F(x) \\ u &= \frac{1}{2} + \frac{1}{\pi} \arctan(x) \\ x &= \tan(\pi(u - 1/2)). \end{aligned}$$

```
n <- 1000
us <- runif(n)
d1 <- tan(pi*(us-0.5))
d2 <- rcauchy(n=n)

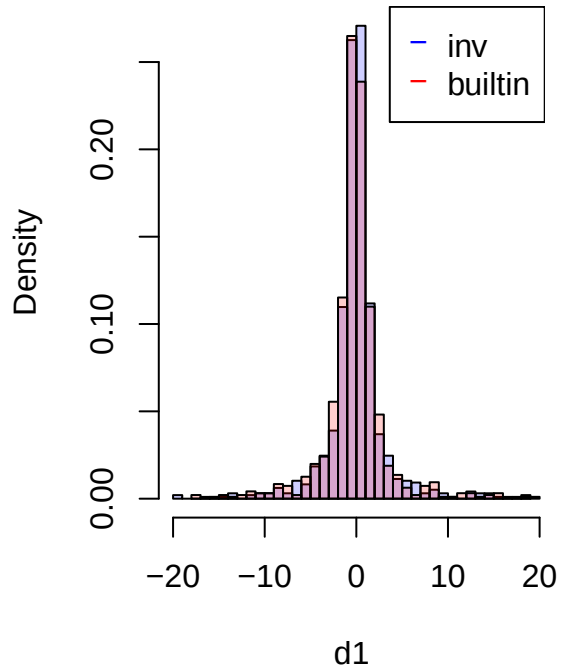
d1 <- d1[d1 > -20 & d1 < 20]
d2 <- d2[d2 > -20 & d2 < 20]

c1 <- rgb(0,0,255, max = 255, alpha = 50, names = "blue")
c2 <- rgb(255,0,0, max = 255, alpha = 50, names = "red")

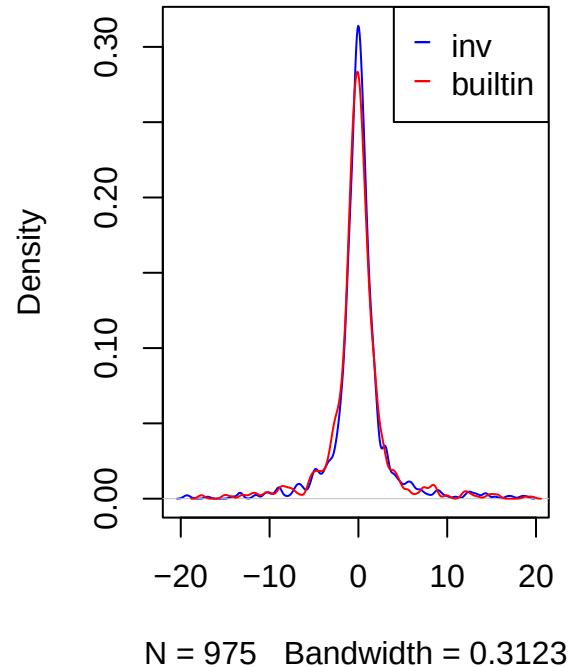
par(mfrow=c(1,2))
hist(d1,col=c1,freq=F,breaks=50,main="inverse-transform method vs built-in")
hist(d2,col=c2,add=T,freq=F,breaks=50)
legend(x="topright",legend=c("inv","builtin"),col=c("blue","red"),pch=c("-", "-"))
plot(density(d1), col="blue",main="density plot")
```

```
lines(density(d2), col="red")
legend(x="topright", legend=c("inv", "builtin"), col=c("blue", "red"), pch=c("-", "-"))
```

inverse-transform method vs built



density plot



Problem 4

Generating 10,000 random variables from $\text{Geometric}(p)$ distribution based off Bernoulli trials.

Solution

A random variable $X \sim \text{Geometric}(p)$ is given by the number of i.i.d. trials needed to have a success where success occurs with probability p .

Thus, we may simulate this distribution with the following R code:

```
# simulate n realizations of geometric(p)
rgeo <- function(n,p)
{
  outcomes <- vector(length=n)
  for (i in 1:n)
  {
    trials <- 0
    while (T)
    {
      trials <- trials + 1
      if (rbinom(1,1,p) == 1)
      {
        break
      }
    }
    outcomes[i] <- trials
  }
}
```

```

    outcomes[i] <- trials
  }
  outcomes
}

```

When we use this function to draw a sample of $n = 10000$ geometrically distributed random variables with $p = 0.2$, we obtain:

```

p <- .2
n <- 10000
sample <- rgeo(n,p)
cat("the mean should be approximate 1/p =", 1/p, " and we obtain a mean of ", mean(sample))

## the mean should be approximate 1/p = 5 and we obtain a mean of 5.009

```

Problem 5

Generate random values from a Standard Half Normal distribution with pdf,

$$f(x) = \frac{2}{\sqrt{2\pi}} e^{-x^2/2}, x > 0.$$

For the candidate pdf, choose the exponential density with rate 1. Verify that your method works via a plot of the true density, and a histogram of the generated values.

Solution

We are given the density of the standard half-normal distribution,

$$\text{dhalfnormal}(x) = \frac{2}{\sqrt{2\pi}} e^{-x^2/2}, x > 0.$$

We model this density with the following R code:

```

# density for standard half-normal
dhalfnormal <- function(x) { 2/sqrt(2*pi)*exp(-x^2/2) }

```

We sample from the exponential distribution $\text{EXP}(\lambda = 1)$, with density g and thus we first find the c satisfying

$$c = \max \left\{ \frac{\text{dhalfnormal}(x)}{\text{dexp}(x|\lambda = 1)} \mid x \in \mathbb{R} \right\},$$

which is found to be approximately $c = 1.315489247$.

We implement the standard half-normal sampler, `rhalfnormal`, using the acceptance-rejection sampling technique with the following R code:

```

# accept-rejection sampling for standard half-normal
# using exp(rate=1)
rhalfnormal <- function(N)
{
  c <- 1.315489247
  xs <- vector(length=N)
  k <- 1
  while (T)
  {
    x <- rexp(n=1)

```



```

if (runif(n=1) < dhalfnormal(x)/(c*dexp(x)))
{
  xs[k] <- x
  k <- k + 1
  if (k == N)
  {
    break
  }
}
}
xs
}

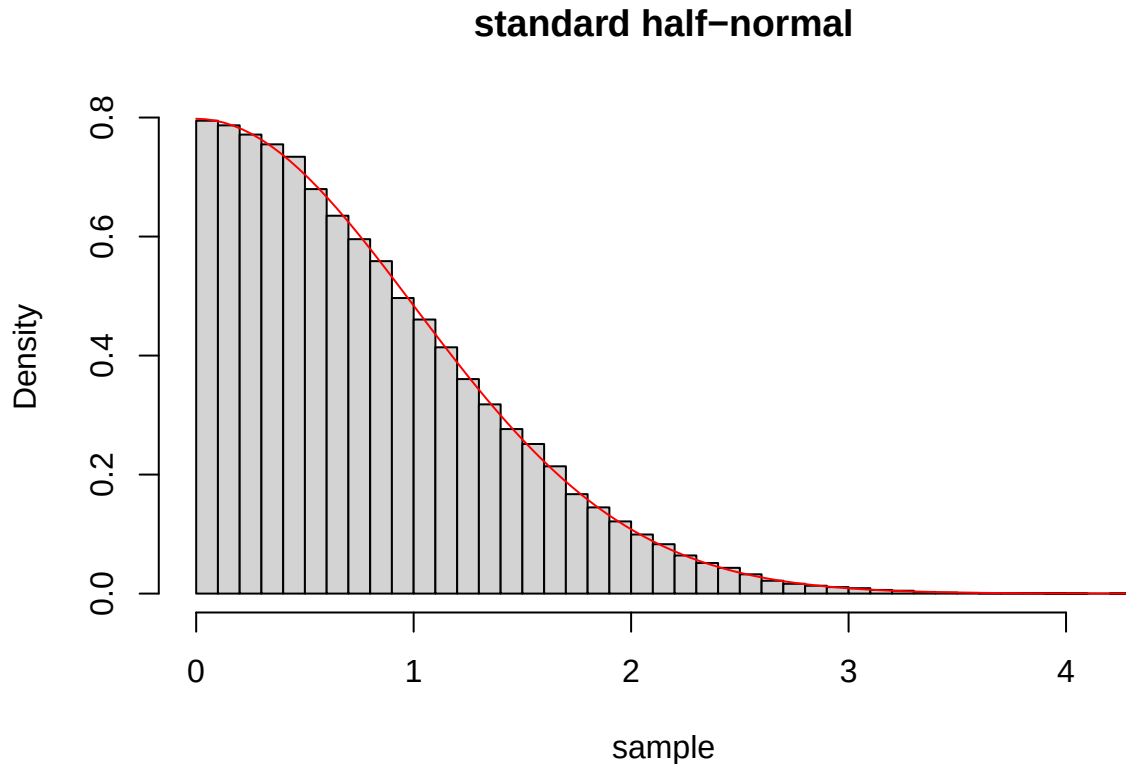
```

We simulate drawing $n = 100000$ samples from the standard half-normal distribution and plotting a histogram of the sample with its density overload in red on top of it with the following R code:

```

n <- 100000
sample <- rhalfnormal(n)
hist(sample,freq=F,breaks=50,main="standard half-normal")
curve(dhalfnormal(x),add=TRUE,col="red")

```



We see that the histogram is compatible with being drawn from the overload density.

Problem 6

Use accept-reject to sample from this bimodal density:

$$f(x) \propto 3e^{-0.5(x+2)^2} + 7e^{-0.5(x-2)^2}$$

The normalizing constant is 25.066. For your proposal $g(\hat{u})$, use a $N(0, 2^2)$ distribution. Verify that your method works via a plot of the true normalized density, and a histogram of the generated values.

Solution

We are given the kernel of the bimodal distribution of interest,

$$\text{ker-bimodal}(x) = 3e^{-0.5(x+2)^2} + 7e^{-0.5(x-2)^2},$$

with the normalizing constant $C = 25.0663$ and thus the pdf for the bimodal is given by

$$\text{dbimodal}(x) = \frac{\text{ker}(x)}{C}.$$

We model these two functions with the following R code:

```
# density for biomodal density
kerbimodal <- function(x) { 3*exp(-0.5*(x+2)^2) + 7*exp(-0.5*(x-2)^2) }
kerbimodal.C <- 25.0663
dbimodal <- function(x) { kerbimodal(x) / kerbimodal.C }
```

We sample from the normal distribution $N(\mu = 0, \sigma^2 = 2^2)$, with density g and thus we first find the c satisfying

$$c = \max \left\{ \frac{\text{ker-bimodal}(x)}{g(x|\mu = 0, \sigma^2 = 2^2)} \mid x \in \mathbb{R} \right\},$$

which is found to be approximately $c = 68.35212$.

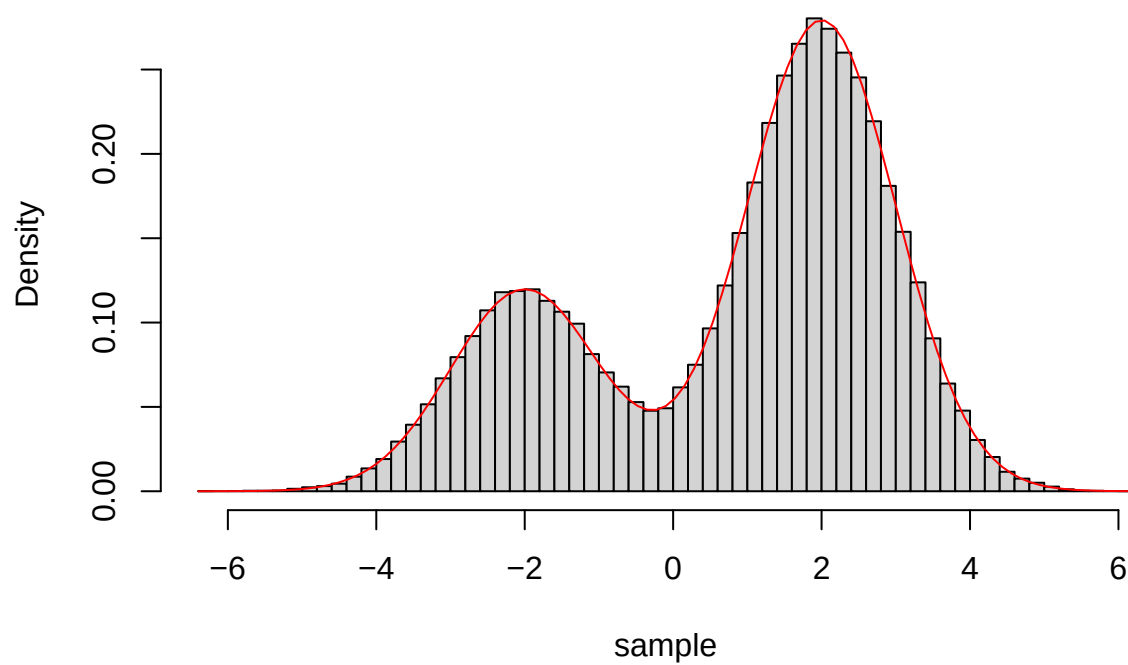
We implement the bimodal sampler, `rbimodal`, using the acceptance-rejection sampling technique with the following R code:

```
# accept-rejection sampling for bimodal distribution with density dbimodal
# using normal(0,2^2).
rbimodal <- function(N)
{
  c <- 68.35212
  xs <- vector(length=N)
  k <- 1
  while (T)
  {
    x <- rnorm(n=1,mean=0,sd=2)
    if (runif(n=1) < kerbimodal(x)/(c*dnorm(x,mean=0,sd=2)))
    {
      xs[k] <- x
      if (k == N)
      {
        break
      }
      k <- k + 1
    }
  }
  xs
}
```

We simulate drawing $n = 100000$ samples from the bimodal distribution and plotting a histogram of the sample with its density overload in red on top of it with the following R code:

```
n <- 100000
sample <- rbimodal(n)
hist(sample,freq=F,breaks=50,main="bimodal")
curve(dbimodal(x),add=TRUE,col="red")
```

bimodal



We see that the histogram is compatible with being drawn from the overload density.